FRACTIONAL KNAPSACK USING GREEDY METHOD

Theory • Algorithm • Example • Complexity • Viva

1. Theory

The **Fractional Knapsack Problem** is an optimization problem in which we are given a set of items, each having a profit and a weight. The objective is to maximize the total profit without exceeding the given knapsack capacity.

It is solved using the **Greedy Method**, where items are selected based on their profit-to-weight ratio. Items with a higher ratio are selected first. Unlike the 0/1 Knapsack problem, Fractional Knapsack allows us to take a fraction of an item if the entire item cannot fit into the remaining capacity.

2. Greedy Strategy

1. Accept the profit and weight of each item.

2. Calculate the profit-to-weight ratio for every item.

3. Sort items in descending order of their ratio.

4. Select items with the highest ratio first.

5. If an item does not fit completely, select the fraction that fits.

6. Calculate and display the maximum profit.

3. Formula

Profit-to-weight ratio: Ratio = Profit / Weight

Profit from a fraction: Fractional Profit = Ratio × Remaining Capacity

4. Example

Consider the following items with a knapsack capacity of 50.

Item	Profit	Weight	Ratio
1	60	10	6
2	100	20	5
3	120	30	4

Selection:
• Item 1: Take completely → Profit = 60.
• Item 2: Take completely → Profit = 100.
• Item 3: Take 20 out of 30 weight → Profit = (120 / 30) × 20 = 80.

Total profit = 60 + 100 + 80 = 240.

5. Complexity Analysis

Time Complexity: O(n log n), due to sorting the items.
Space Complexity: O(n), for storing the items.

6. Applications

• Resource allocation.
• Budget management.
• Cargo loading.

• Investment planning.
• Optimization problems.

7. Main Theory to Remember

Fractional Knapsack uses the **Greedy Method** to maximize profit by selecting items in descending order of their profit-to-weight ratio. It allows taking fractions of items when the remaining capacity is insufficient for a complete item.

8. Viva Question

Q. What is the main difference between Fractional and 0/1 Knapsack?
Fractional Knapsack allows items to be divided, whereas 0/1 Knapsack allows only complete selection or rejection of an item.

9. Conclusion

The Fractional Knapsack problem is efficiently solved using the Greedy Method. By selecting items with the highest profit-to-weight ratio first, we obtain the maximum possible profit within the given capacity.

EXAM-READY LINE

Fractional Knapsack uses the Greedy Method and selects items in descending order of profit-to-weight ratio. Fractions of items are allowed.